# TITLE OF THE ARTICLE FIRST LINE UP TO HERE IF A SECOND LINE IS NEEDED

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*Abstract:* **Contents of the Abstract.**

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*Key words and phrases:* Balanced incomplete block design, Hadamard matrix, nearly balanced incomplete block design, orthogonal array.

## 1. Numerical Study

This section examines the finite-sample performance of the proposed simultaneous confidence band (SCB) procedure and illustrates its use in a longitudinal biomedical application. The simulation study is organized to evaluate simultaneous coverage and band width under several sparse functional data settings that are representative of the theoretical framework. We then apply the method to the Mayo PBC sequential biomarker dataset, where the goal is to conduct simultaneous inference for time-varying bilirubin trajectories and covariate effects.

### 1.1 Simulation

The simulation study is organized into three complementary experiments. The first experiment examines marginal SCB construction for an individual coefficient function in a varying-coefficient model (VCM) with two functional coefficients, using the plug-in SCB of Cao et al. (2018) as the benchmark. The second experiment evaluates the proposed joint SCB for all coefficient functions in a five-dimensional VCM. Since Cao's procedure is designed for a single target function, its joint implementation is obtained by applying a Bonferroni adjustment across coefficient functions. The third experiment considers SCB construction for a mean function under

a Karhunen–Loève data-generating mechanism and compares the proposed method with the B-spline SCB of Cai and Hu (2025). Throughout the simulation study, $n$ denotes the number of subjects, and $m$ denotes the expected number of repeated measurements per subject, with $m_i$ generated uniformly over the integers $\{m-2, \ldots, m+2\}$. The observation times are generated independently as $T_{ij} \sim \text{Uniform}(0, 1)$. All SCBs are evaluated on a common grid of $G = 96$ equally spaced points in $[0.025, 0.975]$, and the nominal significance level is fixed at $\alpha = 0.05$. Each configuration is replicated $R = 1{,}000$ times. For each setting, we report the empirical simultaneous coverage probability, defined as the proportion of Monte Carlo replicates in which the entire target function, or all target coefficient functions in the joint case, is contained in the corresponding band over the evaluation grid, together with the average band width.

Before turning to the individual experiments, we first describe the common implementation choices. Bandwidth selection and undersmoothing are important practical considerations in nonparametric inference, and to keep the numerical comparison focused on the inferential procedure rather than on case-by-case tuning, all simulations use the same two-stage bandwidth selection scheme. The estimation bandwidth $\hat{h}_{\text{est}}$ is selected by leave-one-

subject-out cross-validation,

$$\hat{h}_{\text{est}} = \arg\min_{h \in \mathcal{H}} \frac{1}{N} \sum_{i,j} \{Y_{ij} - Z_{ij}^{\top} \widehat{\beta}_h^{(-i)}(T_{ij})\}^2, \tag{1.1}$$

where $N = \sum_i m_i$, $\widehat{\beta}_h^{(-i)}$ is the pooled local-linear estimator computed after removing subject $i$, and the candidate set is $\mathcal{H} = \text{linspace}(0.04, 0.16, 13)$. The inference bandwidth is then obtained from the common undersmoothing rule

$$\hat{h}_{\text{inf}} = \hat{h}_{\text{est}} \left\{ 1 - \frac{c}{n^{1/2} (\bar{m}/q)^{1/4}} \right\}, \qquad c = 8, \tag{1.2}$$

where $\bar{m}$ is the realized average number of observations per subject and $q$ is the number of coefficient functions in the target model ($q = 2$ in Part 1, $q = 5$ in Part 2, and $q = 1$ in Part 3). The rule is motivated by the usual undersmoothing principle in nonparametric inference: the inference bandwidth is slightly smaller than the prediction-optimal bandwidth, while $\hat{h}_{\text{inf}}/\hat{h}_{\text{est}} \to 1$ as $n \to \infty$ (Hall, 1992; Calonico et al., 2014, 2018). The factor $n^{1/2}$ reflects the subject-level sampling scale of the limiting process, whereas $(\bar{m}/q)^{1/4}$ moderates the amount of undersmoothing according to the within-subject sampling density and the dimension of the coefficient vector, in line with effective-sample-size heuristics for sparse functional data (Yao et al., 2005; Zhang and Wang, 2016). The constant $c = 8$ is fixed once through a pilot calibration and is kept unchanged in all reported experiments.

The plug-in SCB of Cao et al. (2018) is implemented using Silverman pilot bandwidths $b_n = c_n = d_n = 1.06\widehat{\sigma}_t N^{-1/5}$ for the design density, the local design matrix, and the local noise variance, respectively. Its critical value is obtained from $B_{\text{crit}} = 2{,}000$ independent simulations of the limiting Gaussian-process supremum. The B-spline SCB of Cai and Hu (2025) is implemented with cubic B-splines, with the number of interior knots $J_n$ selected by BIC over the range

$$\lceil 0.5 n^{1/4} (\log n)^{1/2} \rceil \le J_n \le \lfloor 2 n^{0.3} \rfloor.$$

The corresponding subject-level sandwich covariance estimator is

$$\widehat{\Sigma} = \widehat{V}^{-1} \left\{ (n N_{\text{avg}}^2)^{-1} \sum_i s_i s_i^\top \right\} \widehat{V}^{-1}, \qquad s_i = \sum_j B(T_{ij}) r_{ij},$$

and the critical value is computed from $B = 500$ multivariate Gaussian draws $Z \sim N(0, \widehat{\Sigma})$ of the standardized spline process. The proposed SCB uses $n_{\text{boot}} = 300$ Gaussian multiplier draws of the standardized supremum process, with leave-one-subject-out residuals evaluated at $\hat{h}_{\text{inf}}$.

*Part 1: single-coefficient VCM.* The first experiment considers the VCM design of Cao et al. (2018), in which the response is generated as

$$Y_{ij} = \beta_0(T_{ij}) + \beta_1(T_{ij}) X_i(T_{ij}) + \zeta_i X_i(T_{ij}). \tag{1.3}$$

Here

$$\beta_0(t) = 5(t - 0.6)^2, \qquad \beta_1(t) = 0.5 + 0.4 \sin\{2\pi(t - 0.5)\},$$

$X_i(\cdot)$ is a centered Gaussian process with pointwise variance $\exp(t)$ and correlation function $\exp(-|\Delta t|/4)$, and $\zeta_i$ is a unit-variance subject-level multiplier. We consider three distributions for $\zeta_i$: $N(0,1)$, Uniform$(-\sqrt{3},\sqrt{3})$, and Laplace$(0,1/\sqrt{2})$. The inferential target is $\beta_1(\cdot)$, and simultaneous coverage requires the SCB to contain $\beta_1(t)$ at all grid points.

The simulation settings are $n \in \{200, 300\}$, $m \in \{5, 10, 15\}$, and the three noise distributions described above, giving 18 configurations in total. Figure 1 shows representative single-run bands, giving a visual sense of the target function, the proposed SCB, and the Cao benchmark before the Monte Carlo summaries are considered. Tables 1 and 2 report empirical simultaneous coverage and average band width.

The proposed SCB exhibits stable calibration across all combinations of $(n, m, \zeta)$. Its empirical coverage ranges from 0.931 to 0.962, with mean coverage 0.945 and mean absolute deviation from the nominal level equal to 0.008. Cao's plug-in SCB is more conservative, with mean coverage 0.996 and no configuration below 0.984, and this should be read in light of its target: it is a sparse-longitudinal-data procedure calibrated through large-sample extreme-value theory. In the present finite-sample settings, especially as $m$ increases, that limiting calibration leads to wider bands than are needed for nominal coverage. The proposed bands are shorter in

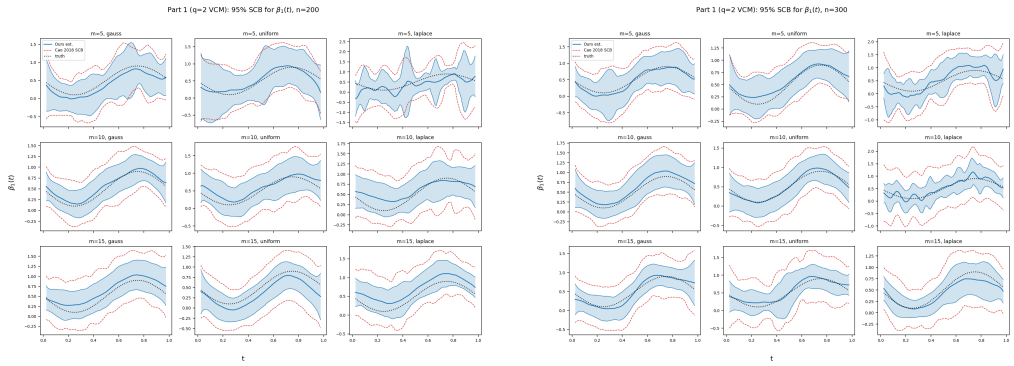


Figure 1: Representative Part 1 bands for the single-coefficient VCM at $n = 200$ and $n = 300$. The figure is used only as a one-run illustration; the tables report Monte Carlo performance.

every setting, with average width ratio 0.64 over all 18 configurations; the gain increases from about 22% shorter at $m = 5$ to about 47% shorter at $m = 15$.

*Part 2: joint VCM inference over five coefficients.* The second experiment moves from a single target curve to joint simultaneous inference in a five-dimensional VCM. The coefficient functions are

$$\beta_0(t) = 1, \qquad \beta_1(t) = 2\sin(2\pi t), \qquad \beta_2(t) = 2(t - 0.5),$$

$$\beta_3(t) = e^t, \qquad \beta_4(t) = \cos(5\pi t).$$

The covariate processes are generated from Karhunen–Loève-type subject-specific loadings. To check robustness beyond a single error law, we use

Table 1: Part 1, $n = 200$: empirical simultaneous coverage and average band width for $\beta_1$. The width ratio is the average width of the proposed band divided by that of Cao's band.

| $n$ | $m$ | noise | Cao cov/wid | **Ours cov/wid** | width ratio |
|---|---|---|---|---|---|
| 200 | 5 | gauss | 0.990/1.385 | **0.953/1.100** | 0.79 |
| 200 | 5 | uniform | 0.984/1.412 | **0.935/1.126** | 0.80 |
| 200 | 5 | laplace | 0.992/1.348 | **0.949/1.066** | 0.79 |
| 200 | 10 | gauss | 1.000/1.408 | **0.947/0.860** | 0.61 |
| 200 | 10 | uniform | 0.999/1.466 | **0.944/0.888** | 0.61 |
| 200 | 10 | laplace | 0.999/1.389 | **0.935/0.841** | 0.61 |
| 200 | 15 | gauss | 1.000/1.462 | **0.937/0.789** | 0.54 |
| 200 | 15 | uniform | 0.999/1.461 | **0.931/0.791** | 0.54 |
| 200 | 15 | laplace | 1.000/1.424 | **0.954/0.764** | 0.54 |

standardized Gaussian, uniform, and Laplace observation noise, with the variance in each case calibrated to yield signal-to-noise ratio 5. The target event is joint simultaneous coverage:

$$\beta_k(t) \in [\hat{L}_k(t), \hat{U}_k(t)] \quad \text{for all } k = 0, \ldots, 4 \text{ and all } t \text{ in the grid.}$$

Accordingly, the multiplier-bootstrap critical value is computed from the

Table 2: Part 1, $n = 300$: empirical simultaneous coverage and average band width for $\beta_1$. The width ratio is the average width of the proposed band divided by that of Cao's band.

| $n$ | $m$ | noise | Cao cov/wid | **Ours cov/wid** | width ratio |
|---|---|---|---|---|---|
| 300 | 5 | gauss | 0.991/1.163 | **0.950/0.871** | 0.75 |
| 300 | 5 | uniform | 0.987/1.156 | **0.947/0.876** | 0.76 |
| 300 | 5 | laplace | 0.996/1.104 | **0.962/0.835** | 0.76 |
| 300 | 10 | gauss | 0.999/1.183 | **0.941/0.698** | 0.59 |
| 300 | 10 | uniform | 0.999/1.175 | **0.931/0.701** | 0.60 |
| 300 | 10 | laplace | 0.999/1.163 | **0.951/0.684** | 0.59 |
| 300 | 15 | gauss | 1.000/1.181 | **0.958/0.633** | 0.54 |
| 300 | 15 | uniform | 0.999/1.201 | **0.942/0.640** | 0.53 |
| 300 | 15 | laplace | 1.000/1.181 | **0.950/0.624** | 0.53 |

supremum over the full index set $(k, t)$.

The settings are $n \in \{200, 300\}$, $m \in \{5, 10, 15\}$, three noise distributions, and $R = 1{,}000$ replications per configuration. Since Cao's plug-in SCB is a marginal, per-target construction, we form a simultaneous benchmark by applying a Bonferroni correction: each coefficient function is as-

Table 3: Part 2, $n = 200$: joint simultaneous coverage over all five coefficient functions. Width is averaged over the five components.

| $m$ | Noise | **Ours cov/wid** | Bonferroni–Cao cov/wid | ratio |
|---|---|---|---|---|
| 5 | Gaussian | **0.967**/1.340 | 1.000/1.617 | 0.83 |
| 5 | Uniform | **0.964**/1.354 | 1.000/1.622 | 0.84 |
| 5 | Laplace | **0.955**/1.322 | 1.000/1.605 | 0.82 |
| 10 | Gaussian | **0.960**/0.849 | 1.000/1.775 | 0.48 |
| 10 | Uniform | **0.962**/0.856 | 1.000/1.782 | 0.48 |
| 10 | Laplace | **0.962**/0.845 | 1.000/1.770 | 0.48 |
| 15 | Gaussian | **0.960**/0.680 | 1.000/1.863 | 0.37 |
| 15 | Uniform | **0.962**/0.683 | 1.000/1.866 | 0.37 |
| 15 | Laplace | **0.949**/0.678 | 1.000/1.861 | 0.36 |

signed a Cao SCB at level $\alpha/p = 0.01$, and the intersection of the five bands is used as the joint SCB.

Tables 3 and 4 show that the proposed joint SCB remains stable over the full grid of sample sizes, sampling densities, and noise distributions. Across the 18 configurations, its empirical coverage ranges from 0.938 to 0.967, with mean coverage 0.954. This is close to the nominal level for

Table 4: Part 2, $n = 300$: joint simultaneous coverage over all five coefficient functions. Width is averaged over the five components.

| $m$ | Noise | **Ours cov/wid** | Bonferroni–Cao cov/wid | ratio |
|---|---|---|---|---|
| 5 | Gaussian | **0.950**/0.969 | 1.000/1.365 | 0.71 |
| 5 | Uniform | **0.967**/0.971 | 1.000/1.364 | 0.71 |
| 5 | Laplace | **0.958**/0.955 | 1.000/1.353 | 0.71 |
| 10 | Gaussian | **0.938**/0.650 | 1.000/1.505 | 0.43 |
| 10 | Uniform | **0.943**/0.655 | 1.000/1.515 | 0.43 |
| 10 | Laplace | **0.942**/0.648 | 1.000/1.503 | 0.43 |
| 15 | Gaussian | **0.941**/0.531 | 1.000/1.578 | 0.34 |
| 15 | Uniform | **0.947**/0.533 | 1.000/1.582 | 0.34 |
| 15 | Laplace | **0.938**/0.531 | 1.000/1.579 | 0.34 |

a band that must cover five coefficient functions simultaneously over the whole time grid. The average width also decreases systematically as the within-subject sampling density increases: averaging over $n$ and the noise laws, the width falls from 1.152 at $m = 5$ to 0.750 at $m = 10$ and 0.606 at $m = 15$.

The Bonferroni-adjusted Cao benchmark also attains simultaneous cov-

erage, but does so conservatively. This is not surprising: a marginal sparse-design band, already based on an extreme-value approximation, is being asked to deliver a joint five-function statement after Bonferroni inflation. Its empirical coverage is 1.000 in all 18 settings, while its mean width is nearly twice that of the proposed band. The relative gain is most visible as $m$ increases: averaged over $n$ and the noise distributions, the proposed joint band uses about 77%, 46%, and 35% of the Bonferroni–Cao width at $m = 5, 10, 15$, respectively. The comparison therefore mainly illustrates the cost of converting a marginal sparse-data procedure into a genuinely joint multi-coefficient band.

*Part 3: mean function inference.* The third experiment considers SCB construction for a mean function under a standard sparse functional data model. The data are generated from

$$Y_{ij} = m(T_{ij}) + \phi(T_{ij})^\top \xi_i + \sigma_\varepsilon \varepsilon_{ij}, \tag{1.4}$$

where

$$m(t) = 1.5 \sin\{3\pi(t + 0.5)\} + 2t^3,$$

$$\phi(t) = \sqrt{2}(\sin 2\pi t, \cos 2\pi t, \sin 4\pi t, \cos 4\pi t).$$

The associated eigenvalues are $(1, 1/2, 1/4, 1/8)$, and $\sigma_\varepsilon = 0.1$. The random variables $\xi_i$ and $\varepsilon_{ij}$ are generated independently from one of three unit-

variance distributions: $N(0,1)$, Uniform$(-\sqrt{3},\sqrt{3})$, or Laplace$(0,1/\sqrt{2})$.

The simulation settings are $n \in \{200, 300\}$, $m \in \{5, 10, 15\}$, and the three error distributions, again giving 18 configurations in total. The proposed local-linear SCB is compared with the B-spline SCB of Cai and Hu (2025).

Tables 5 and 6 show that the proposed SCB is well calibrated in the mean-function setting. Across all 18 configurations, its empirical coverage ranges from 0.941 to 0.965, with mean coverage 0.954 and mean absolute deviation from the nominal level equal to 0.007. The Cai–Hu B-spline SCB is uniformly narrower, with mean width 0.703 compared with 0.849 for the proposed method, and its coverage remains broadly reasonable for a finite-sample comparison. Its mean coverage is 0.926, so the main difference is the familiar efficiency-calibration trade-off: the specialized B-spline construction gives shorter bands, whereas the proposed local-linear procedure stays closer to the nominal simultaneous level while using the same algorithm as in the VCM experiments.

Taken together, the simulations show that the proposed SCB is stable across the settings considered. It performs consistently for a marginal VCM target, continues to work when the inferential target becomes genuinely multi-dimensional and simultaneous, and remains effective when the same

Table 5: Part 3: mean function SCB, comparing the proposed band with the Cai–Hu (2025) B-spline band for $n = 200$.

| $n$ | $m$ | noise | **Ours cov/wid** | Cai–Hu cov/wid |
|---|---|---|---|---|
| 200 | 5 | normal | **0.965/1.115** | 0.919/0.892 |
| 200 | 5 | uniform | **0.961/1.122** | 0.908/0.893 |
| 200 | 5 | laplace | **0.957/1.111** | 0.906/0.889 |
| 200 | 10 | normal | **0.960/0.889** | 0.936/0.743 |
| 200 | 10 | uniform | **0.959/0.890** | 0.931/0.742 |
| 200 | 10 | laplace | **0.953/0.886** | 0.931/0.741 |
| 200 | 15 | normal | **0.951/0.806** | 0.927/0.685 |
| 200 | 15 | uniform | **0.959/0.805** | 0.930/0.686 |
| 200 | 15 | laplace | **0.947/0.800** | 0.941/0.682 |

construction is applied to mean-function inference. This stability, rather than a scenario-specific tuning advantage, is the main numerical message.

Table 6: Part 3: mean function SCB, comparing the proposed band with the Cai–Hu (2025) B-spline band for $n = 300$.

| $n$ | $m$ | noise | **Ours cov/wid** | Cai–Hu cov/wid |
|---|---|---|---|---|
| 300 | 5 | normal | **0.949/0.896** | 0.922/0.732 |
| 300 | 5 | uniform | **0.954/0.901** | 0.923/0.733 |
| 300 | 5 | laplace | **0.958/0.896** | 0.927/0.730 |
| 300 | 10 | normal | **0.941/0.728** | 0.928/0.609 |
| 300 | 10 | uniform | **0.958/0.731** | 0.934/0.608 |
| 300 | 10 | laplace | **0.948/0.730** | 0.920/0.608 |
| 300 | 15 | normal | **0.957/0.661** | 0.923/0.561 |
| 300 | 15 | uniform | **0.943/0.661** | 0.916/0.561 |
| 300 | 15 | laplace | **0.958/0.658** | 0.946/0.559 |

### 1.2 Real data application: Mayo PBC sequential biomarkers

We apply the proposed SCB to the Mayo PBC sequential biomarker dataset (Therneau and Grambsch, 2000), available as `pbcseq` in the R `survival` package. The data arise from the Mayo Clinic randomized D-penicillamine

trial for primary biliary cirrhosis (PBC) reported by Dickson et al. (1985). The full longitudinal record contains 1,945 serial measurements from $n = 312$ patients, with follow-up extending to approximately 13 years. We restrict attention to the first 8.2 years, corresponding to day $\leq 3{,}000$, which retains approximately 92% of the observations. After excluding records with missing bilirubin values, the analysis sample consists of 1,790 observations from all 312 patients.

Figure 4 summarizes the longitudinal design. Measurements are taken at irregular, patient-specific visit times, with a median of 5 observations per patient, a mean of 5.7, and a range of 1–11. The data therefore constitute a sparse longitudinal design, consistent with the sampling regimes considered in the simulation study. The raw bilirubin measurements are strongly right-skewed, with sample skewness 2.88; following standard analyses of this dataset (Therneau and Grambsch, 2000), we analyze log(bilirubin) and rescale time as $t = \text{day}/3000 \in [0, 1]$ for kernel estimation. The treatment arms are nearly balanced, with 158 patients assigned to D-penicillamine and 154 assigned to placebo.

The biological question is not merely whether a smooth curve can be estimated. PBC is a chronic cholestatic liver disease in which progressive bile-duct injury can be reflected in rising bilirubin and alkaline phosphatase,

while impaired liver synthetic function is reflected by lower albumin and longer prothrombin time (Kaplan and Gershwin, 2005; Lindor et al., 2019). Bilirubin is also a central prognostic marker in the Mayo PBC literature (Dickson et al., 1989; Murtaugh et al., 1994). Thus the data allow two linked questions: first, whether the cohort shows a population-level bilirubin elevation over follow-up; and second, whether the time-varying bilirubin trajectory is more closely associated with the randomized D-penicillamine assignment or with contemporaneous markers of cholestasis and hepatic decompensation.

*Mean trajectory.* We begin with the population mean, since it gives the clinical scale of the phenomenon before covariates are introduced. The mean-function model is

$$\log(\text{bili}_{ij}) = m(t_{ij}) + \varepsilon_{ij}. \tag{1.5}$$

For this model, the data-driven bandwidth selection gives $\hat{h}_{\text{est}} = 0.160$ and $\hat{h}_{\text{inf}} = 0.113$. The bootstrap critical value is $\hat{q}_{\text{crit}} = 2.81$, and the average SCB width on the log scale is 0.55. For comparison, the Cai–Hu B-spline SCB selects $J_n = 6$ interior knots and has average width 0.56, so the two real-data mean bands are very similar in overall width; the width ratio of the proposed band relative to the B-spline band is 0.97. Figure 5 displays the estimated trajectory and SCB on both the log scale and the

Table 7: PBC mean bilirubin, back-transformed to mg/dL, with the 95% SCB at selected follow-up times.

| Time | $\widehat{m}(t)$ as bili (mg/dL) | 95% SCB (mg/dL) |
|---|---|---|
| day 135 ($t = 0.04$) | 2.15 | $[1.75, 2.65]$ |
| day 795 ($t = 0.27$) | 2.37 | $[1.83, 3.06]$ |
| day 1545 ($t = 0.52$) | 1.97 | $[1.50, 2.59]$ |
| day 2235 ($t = 0.74$) | 2.02 | $[1.49, 2.74]$ |
| day 2865 ($t = 0.95$) | 1.82 | $[1.24, 2.68]$ |

back-transformed clinical scale, while Table 7 reports selected values on the clinical scale.

The estimated mean trajectory is above the normal reference value of 1.0 mg/dL at the selected time points in Table 7; the lower SCB endpoints are also above this threshold. This provides simultaneous population-level evidence of hyperbilirubinemia in the Mayo PBC cohort, in line with the role of bilirubin as a marker of advanced disease (Dickson et al., 1989; Prince et al., 2004; Corpechot et al., 2008). The curve rises during early follow-up, reaching about 2.4 mg/dL around day 800, and then remains near 2.0 mg/dL through the middle and later portions of the observed window. We

do not interpret the modest flattening and later decline as a typical within-patient recovery pattern. In a longitudinal PBC cohort, the sickest patients may die, receive transplantation, or otherwise leave follow-up earlier; the later cross-sectional mean is therefore computed among a selected set of survivors (Murtaugh et al., 1994; Wulfsohn and Tsiatis, 1997). The SCB is valuable here because it turns this visual impression into a simultaneous statement about the population trajectory over the observed follow-up window.

*Varying-coefficient analysis.* The mean analysis establishes that bilirubin is elevated at the cohort level, but it does not distinguish between treatment assignment and the biological state of the liver. We therefore turn to a varying-coefficient model that includes the randomized D-penicillamine indicator and clinically interpretable markers of decompensation and cholestasis. The outcome remains log(bilirubin), and the covariates are treatment, edema score, albumin, prothrombin time, and log(alkaline phosphatase). Edema, albumin, and prothrombin time are part of the classical Mayo prognostic framework (Dickson et al., 1989; Murtaugh et al., 1994), while alkaline phosphatase is a standard cholestatic marker in PBC assessment and risk stratification (Lindor et al., 2019; Lammers et al., 2014; Carbone et al., 2016). After excluding records with missing alkaline phosphatase,

this analysis uses 1,741 observations from all 312 patients. The non-binary covariates are centered and scaled, so their coefficients are interpreted per one-standard-deviation increase. The model is

$$\begin{aligned}\log(\text{bili}_{ij}) = &\beta_0(t_{ij}) + \beta_1(t_{ij})\text{trt}_i + \beta_2(t_{ij})\text{edema}_{ij}^{\text{std}} \\ &+ \beta_3(t_{ij})\text{albumin}_{ij}^{\text{std}} + \beta_4(t_{ij})\text{protime}_{ij}^{\text{std}} \\ &+ \beta_5(t_{ij})\log(\text{ALP}_{ij})^{\text{std}} + \varepsilon_{ij},\end{aligned} \tag{1.6}$$

where $\text{trt}_i = 1$ denotes assignment to D-penicillamine.

Because inference is required for six coefficient functions over a continuum of time points, we report a joint SCB over all components. The bootstrap supremum is taken jointly over both coefficient index and time, so that the resulting band targets

$$P\{\hat{L}_k(t) \le \beta_k(t) \le \hat{U}_k(t) \text{ for all } k, t\} \approx 0.95.$$

For this joint model, the data-driven bandwidth selection gives $\hat{h}_{\text{est}} = 0.160$ and $\hat{h}_{\text{inf}} = 0.086$. The joint bootstrap critical value is $\hat{q}_{\text{crit}}^{\text{joint}} = 3.59$. For reference, we also report the marginal Cao plug-in SCB, for which the critical constant is $\gamma = 2.82$. The resulting bands are displayed in Figure 6.

The VCM analysis gives a sharper biological interpretation of the elevated mean trajectory. The D-penicillamine band contains zero over the entire follow-up window, consistent with the original trial conclusion that

Table 8: PBC VCM analysis: proposed joint 95% SCB versus the marginal Cao band. The final column records whether the proposed joint band excludes zero over at least one time interval.

| Coefficient | Ours joint wid | Cao wid | width ratio | Joint band excludes 0? |
|---|---|---|---|---|
| $\beta_0$ (intercept) | 0.837 | 0.902 | 0.93 | yes |
| $\beta_1$ (D-penicillamine) | 1.082 | 1.192 | 0.91 | no |
| $\beta_2$ (edema) | 0.613 | 0.689 | 0.89 | yes |
| $\beta_3$ (albumin) | 0.794 | 0.882 | 0.90 | yes |
| $\beta_4$ (prothrombin time) | 1.001 | 0.973 | 1.03 | yes |
| $\beta_5$ (log alkaline phosphatase) | 0.632 | 0.706 | 0.89 | yes |

D-penicillamine did not provide a meaningful clinical benefit in advanced PBC (Dickson et al., 1985; Lindor et al., 2009, 2019). By contrast, the log alkaline phosphatase coefficient is positive over most of the interior time range, indicating that higher cholestatic burden is associated with higher log-bilirubin even after joint simultaneous adjustment for the other clinical markers. This direction agrees with large PBC cohorts showing that alkaline phosphatase and bilirubin jointly summarize cholestatic disease activity and long-term outcome risk (Lammers et al., 2014; Carbone et al., 2016). The edema and prothrombin-time coefficients are also positive over selected intervals, while the albumin coefficient is negative early in follow-up. These signs are biologically coherent and are supported by the Mayo prognostic lit-

erature: edema and prolonged prothrombin time indicate decompensation and impaired synthetic function, whereas higher albumin indicates better preserved synthetic function (Dickson et al., 1989; Murtaugh et al., 1994; Kaplan and Gershwin, 2005). The comparison with the marginal Cao bands is not the main clinical message, but it shows that the proposed joint band remains competitive in width: it is shorter for five of the six coefficients and only slightly wider for prothrombin time, despite protecting the entire six-function statement simultaneously.

Taken together, the real-data analysis answers the motivating clinical problem in two stages. The mean SCB shows that the Mayo cohort has a persistently elevated bilirubin profile on the population scale. The expanded VCM then shows that the time-varying bilirubin trajectory is not detectably shifted by D-penicillamine assignment, but is strongly aligned with cholestasis and hepatic functional status. This is the type of conclusion for which a joint SCB is useful: it distinguishes a null treatment pattern from several biologically interpretable marker associations while preserving simultaneous coverage across functions and time.

**Supplementary Materials**

Contain the brief description of the online supplementary materials.

## Acknowledgements

Write the acknowledgements here.

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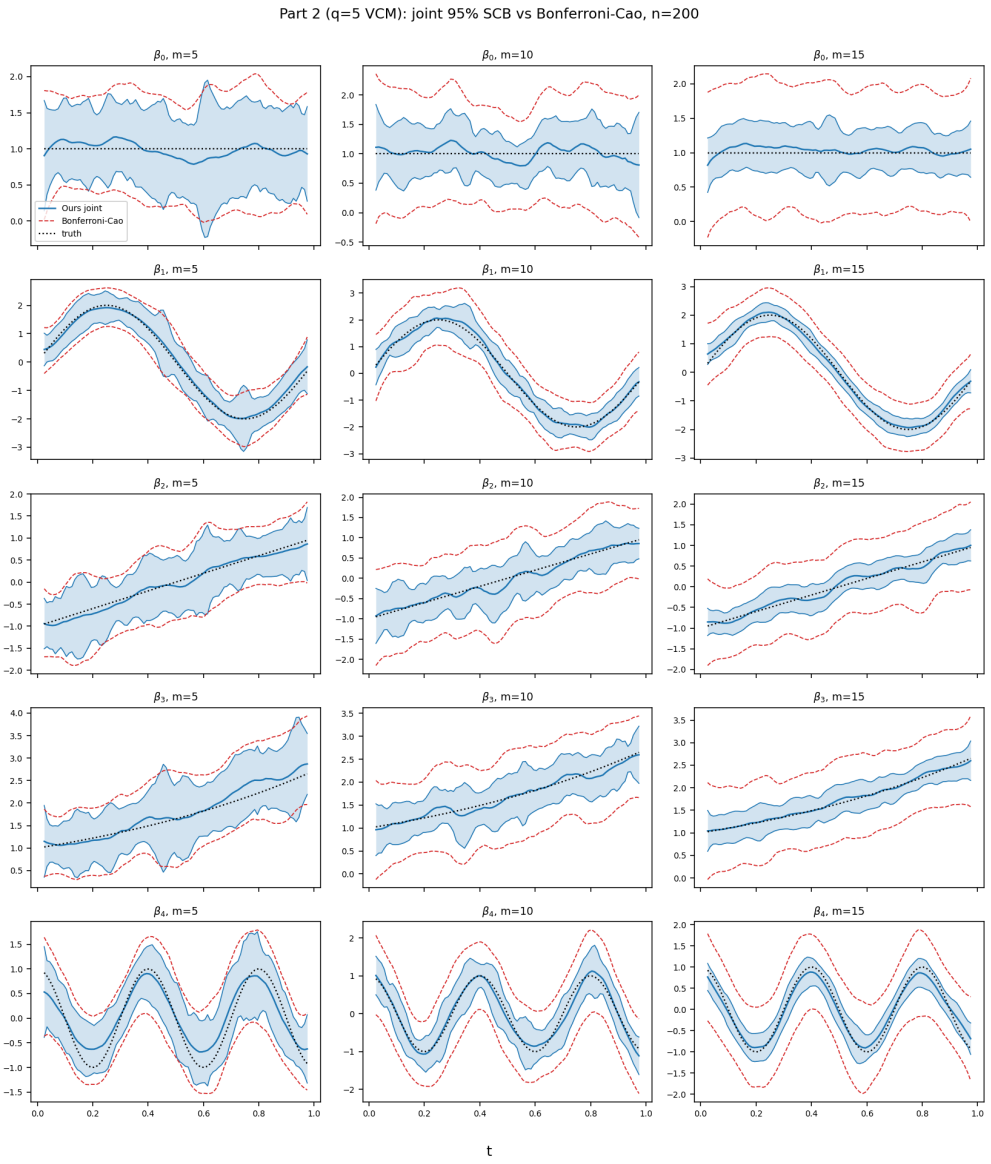



Figure 2: Representative Part 2 joint bands over the five VCM coefficient functions. The display illustrates a single simulated dataset; joint simultaneous coverage is evaluated in Tables 3 and 4.

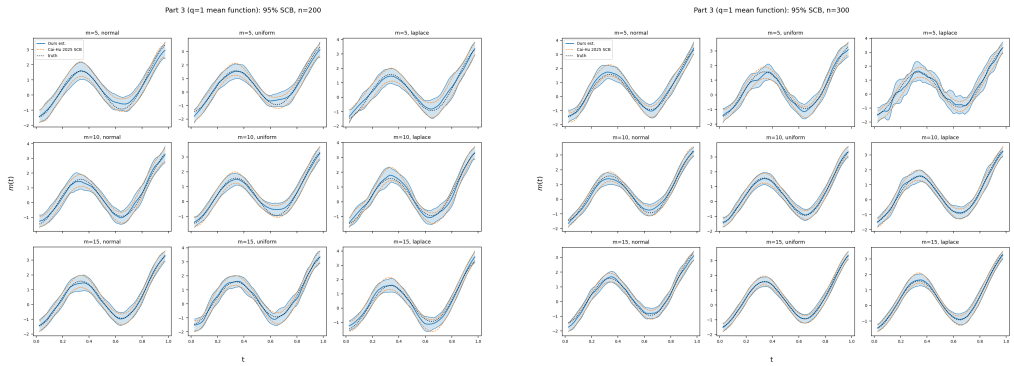



Figure 3: Representative Part 3 bands for mean-function inference at $n = 200$ and $n = 300$. As in the preceding figures, these are single-run illustrations, while Tables 5–6 summarize repeated-sampling behavior.

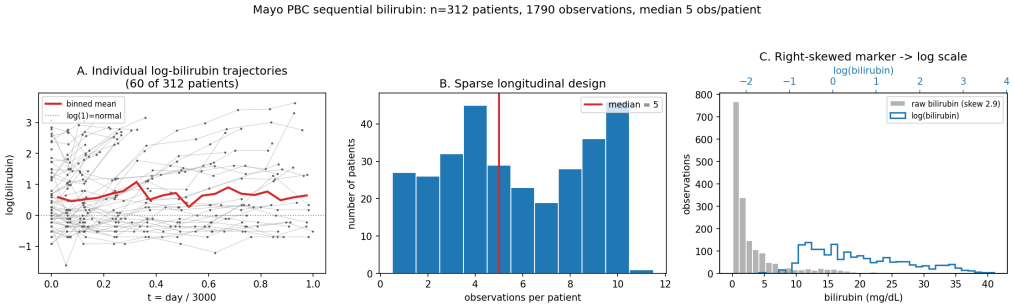



Figure 4: Mayo PBC design illustration: individual log-bilirubin trajectories, the distribution of repeated measurements per patient, and the marginal distribution of raw bilirubin.

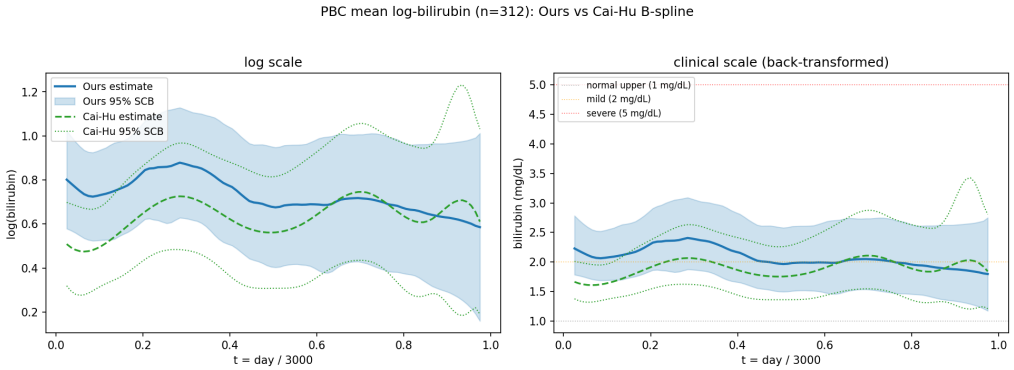



Figure 5: PBC mean bilirubin trajectory with simultaneous confidence bands on the log scale and on the back-transformed clinical scale.

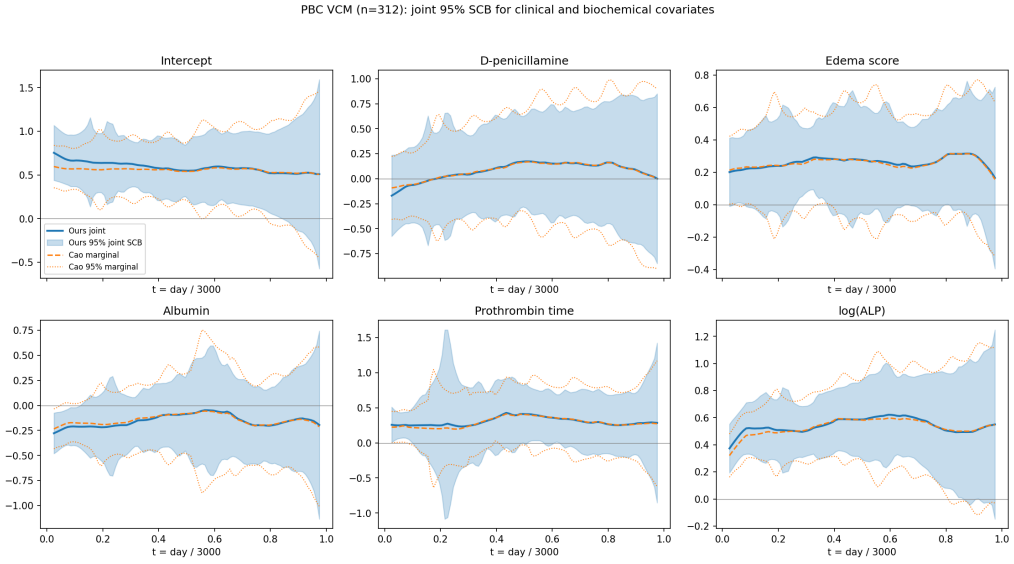



Figure 6: PBC varying-coefficient analysis: joint 95% SCB for treatment, edema, albumin, prothrombin time, and log alkaline phosphatase effects.